

SOFTWARE ENGINEERING PROJECT PLAN

Online Flashcard System for Study

Project Title: Online Flashcard System for Study

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Technologies: JavaFX, MariaDB

Document Purpose: Sprint 1 Project Plan Submission

1. PROJECT OVERVIEW

Problem Summary: Students struggle to keep their study materials organized and often lack efficient ways to practice active recall before exams. Teachers also lack a direct, simple system to distribute pre-made study flashcards to their classes.

Intended Audience/Users: High school and university students looking for a simple study tool, and teachers who want to share study sets with their classes.

Main Features/Components:

- **Student Features:** Create custom flashcards, build practice quizzes, flip through flashcards to study, and view progress statistics.
- **Teacher Features:** Upload subject-specific study materials and flashcard sets for students.
- **Data Management:** Database storage for accounts, cards, decks, and statistics.

2. PROJECT OBJECTIVES

- Develop a functional desktop application using JavaFX and MariaDB by the end of the project timeline.
- Implement basic flashcard creation, editing, and interactive quiz modes for students.
- Enable teachers to upload and share study materials seamlessly.
- Track user study statistics accurately to display student reviewing progress.
- Ensure the user interface is simple, fast, and easy to use without prior training.

3. SCOPE AND DELIVERABLES

In Scope:

- Student creation and management of flashcards and quizzes.
- Teacher upload and management of class flashcard sets.
- Progress tracking and review statistics for students.
- JavaFX Graphical User Interface (GUI).
- MariaDB database setup for persistent data storage.

Out of Scope:

- Mobile app versions (iOS/Android).
- Complex AI-based flashcard auto-generation.
- Online payment or monetization systems.

5. RESOURCE ALLOCATION

Team Members and Roles:

- **Frans Huhta-Koivisto:** Scrum Master & Software Developer
- **Rafi Nadri:** Software Developer (UI & Backend)
- **Artturi Maanoja:** Software Developer (Backend)

Software, Hardware, and Tools:

- **IDE:** IntelliJ IDEA / Eclipse
- **Programming Language & Framework:** Java 17+, JavaFX
- **Database:** MariaDB
- **Project Management & Version Control:** Jira / Trello, Git, GitHub
- **External Resources or Support:** Moodle/Oma course guidelines and JavaFX/MariaDB documentation.

6. RISK MANAGEMENT

Description	Likelihood	Impact	Mitigation Strategy
Database Connection Failure	Medium	High	Create a solid data access layer (DAO) early and test DB connection scripts thoroughly.
JavaFX UI Compatibility Issues	Low	Medium	Stick to standard JavaFX components and test on team computers regularly.
Schedule Delays / Time Constraints	Medium	Medium	Use Jira/Trello to track tasks in small chunks and prioritize core features first.

7. TESTING AND QUALITY ASSURANCE

- **Unit Testing:** Testing individual backend Java methods using JUnit.
- **Integration Testing:** Verifying communication between JavaFX controllers and MariaDB.
- **User Acceptance Testing (UAT):** Manual testing of student and teacher user flows.
- **Criteria for Success:** Zero critical bugs, smooth navigation, and reliable saving/retrieval of study sets.